

OS Lab

Skill-7

Process Synchronization and Child Process Monitoring Using waitpid().

To Retrieve PATH Variable, Parse Search Directories, Locate Executables, Verify Permissions, Handle Missing Commands, Test Command Resolution.

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;
    int status;

    printf("Parent process started. PID = %d\n", getpid());

    pid = fork();

    if (pid < 0)
    {
        perror("fork failed");
        return 1;
    }

    if (pid == 0)
    {
        // Child process
        printf("Child process started. PID = %d\n", getpid());

        printf("Child is doing some work...\n");

        sleep(3);

        printf("Child process completed.\n");

        exit(42);
    }
}
```

```

else
{
    // Parent process
    printf("Parent created child with PID = %d\n", pid);

    printf("Parent is waiting for child...\n");

    pid_t result = waitpid(pid, &status, 0);

    if (result == -1)
    {
        perror("waitpid failed");
        return 1;
    }

    printf("Parent detected child termination.\n");

    if (WIFEXITED(status))
    {
        printf("Child exited normally.\n");
        printf("Child exit status = %d\n",
            WEXITSTATUS(status));
    }
    else if (WIFSIGNALED(status))
    {
        printf("Child was terminated by a signal.\n");
    }
}

printf("Parent process finished.\n");

return 0;
}

```

```

lahari@DESKTOP-AIC33F7:~$ cd ~/OSLab/shell_lab
lahari@DESKTOP-AIC33F7:~/OSLab/shell_lab$ nano waitpid_demo.c
lahari@DESKTOP-AIC33F7:~/OSLab/shell_lab$ gcc waitpid_demo.c -o waitpid_demo
lahari@DESKTOP-AIC33F7:~/OSLab/shell_lab$ ./waitpid_demo
Parent process started. PID = 1206
Parent created child with PID = 1207
Parent is waiting for child...
Child process started. PID = 1207
Child is doing some work...
Child process completed.
Parent detected child termination.
Child exited normally.
Child exit status = 42
Parent process finished.

```

```

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/stat.h>

```

```

#define MAX_PATH_LENGTH 4096

```

```

int is_executable(const char *path)
{
    struct stat file_info;

    // Check whether file exists
    if (stat(path, &file_info) != 0)
    {
        return 0;
    }

    // Check whether it is a regular file
    if (!S_ISREG(file_info.st_mode))
    {
        return 0;
    }

    // Check execute permission
    if (access(path, X_OK) != 0)
    {
        return 0;
    }

    return 1;
}

```

```

}

int main(int argc, char *argv[])
{
    char *path_variable;
    char *path_copy;
    char *directory;

    if (argc != 2)
    {
        printf("Usage: %s <command>\n", argv[0]);
        return 1;
    }

    char *command = argv[1];

    // If command contains '/', check it directly
    if (strchr(command, '/') != NULL)
    {
        if (is_executable(command))
        {
            printf("Executable found: %s\n", command);
            return 0;
        }
        else
        {
            printf("Command is not executable or does not exist: %s\n",
                command);
            return 1;
        }
    }

    // Retrieve PATH variable
    path_variable = getenv("PATH");

    if (path_variable == NULL)
    {
        printf("PATH variable not found.\n");
        return 1;
    }

    // Make a copy because strtok modifies the string
    path_copy = malloc(strlen(path_variable) + 1);

```

```

if (path_copy == NULL)
{
    perror("malloc");
    return 1;
}

strcpy(path_copy, path_variable);

// Parse PATH directories
directory = strtok(path_copy, ":");

while (directory != NULL)
{
    char full_path[MAX_PATH_LENGTH];

    snprintf(full_path,
             sizeof(full_path),
             "%s/%s",
             directory,
             command);

    printf("Checking: %s\n", full_path);

    // Verify executable
    if (is_executable(full_path))
    {
        printf("\nExecutable found!\n");
        printf("Command: %s\n", command);
        printf("Path: %s\n", full_path);

        free(path_copy);

        return 0;
    }

    directory = strtok(NULL, ":");
}

printf("\nCommand not found: %s\n", command);

free(path_copy);

return 1;
}

```

```

lahari@DESKTOP-AIC33F7:~/OSLab/shell_lab$ ls
mini_shell  mini_shell.c  shell_parser  shell_parser.c  waitpid_demo  waitpid_demo.c
lahari@DESKTOP-AIC33F7:~/OSLab/shell_lab$ nano path_resolver.c
lahari@DESKTOP-AIC33F7:~/OSLab/shell_lab$ gcc path_resolver.c -o path_resolver
lahari@DESKTOP-AIC33F7:~/OSLab/shell_lab$ ./path_resolver ls
Checking: /usr/local/sbin/ls
Checking: /usr/local/bin/ls
Checking: /usr/sbin/ls
Checking: /usr/bin/ls

Executable found!
Command: ls
Path: /usr/bin/ls
lahari@DESKTOP-AIC33F7:~/OSLab/shell_lab$ ./path_resolver gcc
Checking: /usr/local/sbin/gcc
Checking: /usr/local/bin/gcc
Checking: /usr/sbin/gcc
Checking: /usr/bin/gcc

Executable found!
Command: gcc
Path: /usr/bin/gcc
lahari@DESKTOP-AIC33F7:~/OSLab/shell_lab$ ./path_resolver pwd
Checking: /usr/local/sbin/pwd
Checking: /usr/local/bin/pwd
Checking: /usr/sbin/pwd
Checking: /usr/bin/pwd

Executable found!
Command: pwd
Path: /usr/bin/pwd

```

```

Path: /usr/bin/pwd
lahari@DESKTOP-AIC33F7:~/OSLab/shell_lab$ ./path_resolver pwd
Checking: /usr/local/sbin/pwd
Checking: /usr/local/bin/pwd
Checking: /usr/sbin/pwd
Checking: /usr/bin/pwd

Executable found!
Command: pwd
Path: /usr/bin/pwd
lahari@DESKTOP-AIC33F7:~/OSLab/shell_lab$ ./path_resolver abc123
Checking: /usr/local/sbin/abc123
Checking: /usr/local/bin/abc123
Checking: /usr/sbin/abc123
Checking: /bin/abc123
Checking: /usr/games/abc123
Checking: /usr/local/games/abc123
Checking: /usr/lib/abi/abc123
Checking: /mnt/c/Program Files/Common Files/Oracle/Java/javapath/abc123
Checking: /mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath/abc123
Checking: /mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath/abc123
Checking: /mnt/c/Program Files/Eclipse Adoptium/jdk-17.0.16-hotspot/bin/abc123
Checking: /mnt/c/WINDOWS/system32/abc123
Checking: /mnt/c/WINDOWS/abc123
Checking: /mnt/c/WINDOWS/System32/WinSxS/abc123
Checking: /mnt/c/WINDOWS/System32/WindowsPowerShell/v1.0/abc123
Checking: /mnt/c/WINDOWS/System32/OpenSSH/abc123
Checking: /mnt/c/Program Files/abc123
Checking: /mnt/c/Program Files/Java/jdk-25/abc123
Checking: /mnt/c/Program Files/Git/cmd/abc123
Checking: /mnt/c/Program Files/nodejs/abc123
Checking: /mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin/abc123
Checking: /mnt/c/Users/ADMIN/AppData/Local/Programs/Python/Python314/Scripts/abc123
Checking: /mnt/c/Users/ADMIN/AppData/Local/Programs/Python/Python314/abc123
Checking: /mnt/c/Users/ADMIN/AppData/Local/Programs/Python/Launcher/abc123
Checking: /mnt/c/mys64/ucrt64/bin/abc123
Checking: /mnt/c/Users/ADMIN/AppData/Local/Programs/Microsoft VS Code/bin/abc123
Checking: /mnt/c/Program Files/JetBrains/PyCharm 2026.1/bin/abc123
Checking: /mnt/c/Program Files/JetBrains/DataGrip 2026.2.1/bin/abc123
Checking: /mnt/c/Users/ADMIN/AppData/Roaming/npm/abc123
Checking: /snap/bin/abc123

Command not found: abc123
lahari@DESKTOP-AIC33F7:~/OSLab/shell_lab$ echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/abi:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Program Files/Eclipse Adoptium/jdk-17.0.16-hotspot/bin:/mnt/c/WINDOWS/system32:/mnt/c/WINDOWS/System32/WinSxS:/mnt/c/WINDOWS/System32/WindowsPowerShell/v1.0:/mnt/c/WINDOWS/System32/OpenSSH:/mnt/c/Program Files/abc123:/mnt/c/Program Files/Java/jdk-25:/mnt/c/Program Files/Git/cmd:/mnt/c/Program Files/nodejs:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin:/mnt/c/Users/ADMIN/AppData/Local/Programs/Python/Python314/Scripts:/mnt/c/Users/ADMIN/AppData/Local/Programs/Python/Python314:/mnt/c/Users/ADMIN/AppData/Local/Programs/Python/Launcher:/mnt/c/mys64/ucrt64/bin:/mnt/c/Users/ADMIN/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Program Files/JetBrains/PyCharm 2026.1/bin:/mnt/c/Program Files/JetBrains/DataGrip 2026.2.1/bin:/mnt/c/Users/ADMIN/AppData/Roaming/npm:/snap/bin
lahari@DESKTOP-AIC33F7:~/OSLab/shell_lab$

```